Bernstein Center for
Computational Neuroscience
Berlin

Lab Rotation

Lab Rotation

im Master-Studiengang Computational Neuroscience
Master Program Computational Neuroscience

1. Anmeldung / Application

Name des Studierenden/student's name:	Shokoofehsadat Hosseini
Matrikelnummer/Matriculation number:	506378
Name des Betreuers: supervisor's name:	Professor Dr. Klaus Obermayer
Adresse des Betreuers: supervisor's address:	Marchstrasse 23, 10587 Berlin
Titel der Lab Rotation: title of lab rotation:	Linking Scene Graph Generation Models with Human Visual Attention Graphs

BCCN-Betreuer (**für externe Lab Rotations**):
name of BCCN supervisor (**for external Lab Rotations**):

Kurzbeschreibung angefügt:
short description attached: ☑

Schwerpunkt/Focus experimentell/experimental ☐ theoretisch/theoretical ☑

Zeitraum/Datum (max. 7 Wochen):
Time period (maximum of 7 weeks): 24.11.2025 - 31.01.2026 (including Christmas break)

Project proposal und Zeitplan angenommen, „Guidelines" (Seite 3) zur Kenntnis genommen:
Project proposal and work schedule approved, „Guidelines" (p. 3) read:

06.10.2025

Datum, Unterschrift des Antragstellers
date, signature student

9.10.2025

Datum, Unterschrift des BCCN-Betreuers
date, signature head supervisor

Datum, Unterschrift des zuständigen Prüfungsausschusses
date, signature head of examination committee

Technische Universität Berlin

Science of Intelligence

Neural Information Processing Group

P.I.: Prof. Dr. Klaus Obermayer

LAB ROTATION OFFER

Project description:

This project investigates the alignment between scene graph generation models and human visual attention patterns. The student will construct graph-based representations of human eye movements and compare them to predicted scene graphs for the same visual scenes. By analysing the structural and semantic correspondences between these graphs, the student explores how closely predicted relations by computer vision models reflect human perceptual organisation. Such correspondence could inform improvements in scene graph generation models, and conversely, predicted relations might improve human scanpath predictions.

This lab rotation requires a **7-weeks commitment**, amounting to **270 hours** of work from **24.11.2025** until **31.01.2025**.

The student will be required to:

- Conduct literature research on the topic
- Analyse human experimental data and build graph-based representations of visual attention
- Predict scene graphs for the same visual scenes
- Analyse the structural and semantic correspondences between these graphs

Required skills:

- Good computational/programming skills – recommended
- Good proficiency in English
- Experience in analysing human data
- Experience in computer vision

The **Neural Information Processing Group** is supervised by **Prof. Dr. Klaus Obermayer**. The group focuses on understanding the principles that govern information processing in biological systems. This involves studying how the brain performs computations, as well as applying the strategies used by biological systems to advance machine learning applications.
The Cluster of Excellence **Science of Intelligence** in Berlin brings together researchers from diverse fields to uncover the fundamental principles of intelligence—across biological and artificial agents, and individuals as well as collective systems—and to translate these insights into new technologies and ethical frameworks for intelligent behaviour. The focus lies on examining and building intelligent behaviour in order to understand it.

Contact: Ole Hall (direct supervisor and PhD student): o.hall@tu-berlin.de